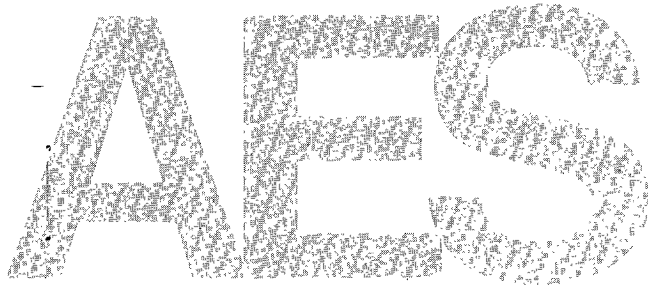


NARROW BAND IMPULSE TESTING OF ACOUSTICAL SYSTEMS

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NARROW BAND IMPULSE TESTING FOR ACOUSTICAL SYSTEMS

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Abstract:

A properly windowed tone burst is used to evaluate the dynamic behavior of a loudspeaker within narrow frequency bands. A clear visual presentation is obtained simultaneously of the speaker's frequency response and the transient behavior. Phase and signal delay information can be derived from the received test signal. The influence of room reflections can be eliminated to obtain anechoic data.

PRESENT MEASUREMENT TECHNIQUES AND THEIR LIMITATIONS

A. Frequency Response

It is standard procedure to measure the steady-state frequency response of acoustical transducers, such as microphones and loudspeakers. Typically, only the amplitude response is measured but with the greater availability of Fourier Analyzers today, it is no longer difficult to also measure the phase response. Acoustical measurements are easily upset by reflections from nearby surfaces, therefore anechoic chambers have been used to obtain reproducible test data for magnitude and phase. Again, a Fourier Analyzer can greatly reduce the need for an anechoic chamber when a short duration pulse of about 10 μ sec is used to excite a speaker and a Fourier transform is performed on the received pulse. [1] If the input to the analyzer is gated so that it blocks reflections which arrive after the

initial output from the speaker, then an anechoic measurement is being performed. The low frequency limit of this test approach is directly related to the time that the gate is open before an echo arrives. Large rooms are needed to extend the frequency range to below a few hundred hertz. Similarly, anechoic chambers have to be large to make low frequency measurements with continuous sine waves.

Yet another approach to discriminate against reflections is the use of a sweeping source and a tracking narrow band receiver. If the sweep rate is sufficiently high then any reflected signal travelling a different path length from the direct signal will be rejected by the receiver because it is detuned relative to it. [2]

Measurement of the steady-state frequency response is important because there exists a large body of information and experience that correlates the frequency domain information with the subjectively perceived sound. This correlation is not always adequate, particularly when it is only based on magnitude response information. Additional phase information is helpful, but it is in general not easily interpreted.

Most program material is transient in nature. The relationship of a measured steady-state frequency response to its effect upon a transient signal is not obvious, even though all the information is contained in the magnitude and phase response necessary to predict a system's output waveform for any given input signal.

B. Transient Response

The difficulty in relating steady-state frequency response to transient response in the time domain is aggravated in multi-driver loudspeaker systems. These are almost always non-minimum phase systems so that no conclusion can be drawn from their frequency response curve as to their waveform distortion. For this reason, transient type signals are often used to evaluate their performance. A short duration impulse will excite the system at all frequencies but again it is difficult to interpret the impulse response in the time domain [3,4,9] The shape of the impulse response is strongly determined by the low frequency cut-off characteristics of the system and by any dominant resonances. This masks low level effects at other frequencies.

A square wave as test signal is more frequency selective; its spectral content consists of odd order harmonics with decreasing amplitude. It is a wideband signal and the response to it is very phase sensitive. It is not clear how much the system response can deviate from the original square wave before being subjectively audible. Signals with considerably distorted waveforms due to phase shift sound subjectively unchanged [5]. The square wave therefore has insufficient correlation to subjective perception.

It is desirable to use a test signal that selectively probes the frequency response and at the same time, gives information about the transient behavior. The tone burst has its spectral energy centered at the sinewave frequency.

If the length of the burst T_B is long enough then the system response to it contains visual information about the transient, and the steady-state behavior of the system in the vicinity of the burst frequency. Both time domain and frequency domain information is obtainable from a CRT display of the system response [8]. Furthermore, if the time between bursts T_R is long enough so that echoes due to reflections have decayed before the start of a new burst, and if the first echo does not overlap with the system response, then anechoic data can be taken.

Test equipment is available which uses a gated receiver to pick out that portion of the received burst, which is of interest and measures the peak amplitude and phase [6]. Each burst consists of an integral number of cycles. The burst length T_B is approximately constant. As the frequency increases, each burst will contain an increasing number of sine wave cycles. Considering the frequency spectrum of such a signal one finds that its envelope, which has a $\sin x/x$ shape, is independent of the burst frequency. Therefore, at low frequencies a larger percentage of the system bandwidth is probed than at high frequencies.

In acoustics, it is desirable to use a constant percentage bandwidth analysis, such as octave or third octave, since it correlates more with subjective perception. By holding the number of cycles in the burst constant, as their frequency changes, the spectrum of the burst is spread a constant percentage around the burst frequency. This is the "counted" tone burst. The burst length T_B is inversely proportional to frequency.

As indicated before, it is desirable to have a test signal that would give a measure of both the steady-state frequency response and the transient behavior of the acoustic system under test. If the counted burst is long enough in duration so that the system can settle to its steady-state response, then one of the desired parameters can be determined from a CRT display, for example. The transient information is non-optimum though. The envelope of the received burst is strongly influenced by the amplitude and phase response of the system under test at frequencies far below the burst frequency. The reason for this is the slow roll-off of the frequency shifted $\sin x/x$ spectrum envelope at low frequencies. The burst, with its rectangular envelope, is exciting the step response of the system but shifted in frequency. Thus, low frequency transients can easily mask the behavior in the vicinity of the burst frequency. Therefore, the rectangular burst response is difficult to interpret.

DYNAMIC RESPONSE

The envelope of the counted tone burst can be modified to reduce the low frequency spectral content. Building up each burst gradually allows the system to respond with minimum overshoot or ringing. Similarly, turning off the burst more gradually, than a step function, lets the system decay without a sharp turn-off transient. Of course, if turn-on and turn-off of the burst are slower than the system attack and decay times, no transient information will be contained in the received signal but only the steady-state response.

There are numerous envelopes that could be used for the counted burst. In the frequency domain, the effect of using an envelope other than the rectangular window is to widen the main lobe of the spectrum and to cause the side lobes to fall off more rapidly. All the different window functions used with discrete Fourier transform techniques could be applied to the counted burst. [7] The Hanning window or $\cos^2$ envelope has been found adequate for these tests.

There is the question of how many cycles the burst should consist of. Using a large number of cycles will hide the transient behavior. The maximum amplitude of the received signal is then proportional to the steady-state frequency response and the phase response is contained in the zero crossings.

A short burst of few cycles will emphasize the transient behavior. The system response though, may not reach full steady-state amplitude. A short duration burst is advantageous in discriminating against echoes. This is important at low frequencies because the burst length T_B for a constant number of cycles increases as the burst frequency is lowered. Therefore, there is less time separation between the direct signal and the first echo. This limits the low frequency range, similar to the maximum window length limit for the Fourier Analyzer. With increasing frequency though, the time discrimination of the counted burst against echoes improves; an important factor for reproducible measurements.

As a compromise between transient and steady-state sensitivity and a low frequency limit of a few hundred hertz

(depending on room size) a burst length of 5 cycles was chosen. A Hanning window, as envelope, gives a mainlobe width which corresponds roughly to third octave analysis.

This test signal gives a positive indication of localized resonances by the shape of the envelope decay. The peak amplitude of the received signal is proportional to the steady-state frequency response, unless the system had insufficient time to build up to it's full response. For this reason the shaped burst test is called dynamic, being a measure of the dynamic behavior of the system when subjected to the specific test signal. The response straddles the conventional frequency and time domain representations and contains useful information about both.

A dynamic frequency response curve is taken by measuring the peak-to-peak amplitude. Resonances are readily seen in the tail of the burst. The zero crossings of the received signal, when compared to the input burst, are a measure of the signal delay. This is important information when trying to combine the outputs from a multi-driver loudspeaker. For example, the time between the fifth zero crossing of the transmitted and the received signal can be measured as a function of burst frequency. This time is a combined measure of the phase delay and the group delay of the system in response to that specific test signal. Of course, all this data could be derived from a proper processing of a system's impulse response. The advantage of the dynamic test lies in the simplicity of the required test equipment. It gives meaningful and easily

interpreted data, which correlate well with subjective perception. As all signals, which contain information, are time varying and transient a system must properly respond to that type of signal, regardless how flat its steady-state amplitude response.

The shaped burst concentrates the energy into a narrow frequency band and its transient information is therefore frequency selective and not as easily masked by system peculiarities of some different frequency. It is not a contradiction to call this a "narrow band impulse test".

The shape of the burst envelope guarantees that there is only a single maximum in the received signal. A peak detecting receiver, therefore, requires no gate that has to be placed over the received signal in order to measure its amplitude.

Discrimination against echoes improves automatically with increasing frequency. The lower frequency limit for freefield measurements is reached when the difference in propagation time between the direct signal and the reflected signal equals the burst length. Below this frequency, the echo will introduce errors depending on its relative amplitude and phase. Echoes can be easily monitored on an oscilloscope because of the distinctive shape of the burst. The repetition rate of the burst has to be low enough (10 Hz) so that all echoes have decayed sufficiently before the next burst occurs. The low duty cycle is also convenient for high power testing of speakers, especially tweeters, in order to determine their excursion

non-linearities and limits.

Finally, it has to be mentioned that the sound of the shaped burst is unobtrusive and pleasant in comparison to rectangular bursts, third octave noise, pink noise, swept sine waves or warble tones.

TEST SYSTEM

The output from a conventional, tunable sine wave generator is applied to a programmable attenuator, which is used to modulate the sine wave. The attenuation changes at each zero-crossing of the sine wave and is constant for each half cycle. The discontinuity in the slope of the output signal at the zero crossing causes odd order harmonics of the burst frequency. This method of generating a specific envelope is only an approximation, but it still gives the desired spectral distribution. The harmonics limit the dynamic range when measuring low frequency cut-off attenuation of a system. In an improved version of the burst generator, the entire wave form might be stored in read-only memory in digitized form. The memory content would be read out at different clock rates, in order to obtain different burst frequencies at the output of a digital to analog converter. A simple reparate generator determines the time between bursts.

The received burst is analyzed by observation on an oscilloscope. The amplitude is measured with a peak detector, after full wave rectification of the incoming signal. Full wave rectification also enhances the visibility of the burst

envelope. Furthermore, the signal is logged before displaying it on the CRT for a greater visual amplitude range.

The burst generator has a calibrated timer, which gets triggered at the fifth zero crossing of each burst. It is used to measure signal delay. A pulse is injected into the log-amplifier of the receiver when the timer resets. By adjusting the reset time, this pulse can be made to coincide with the fifth zero crossing of the received signal, as seen on the scope, and thus a measure of signal delay is obtained.

With a more refined test system, data could be taken on a swept basis but even using point by point measurements this test system has been found to be extremely powerful for development work on speaker systems and for driver quality evaluations.

SUMMARY

A shaped, counted burst has been used for dynamic testing of loudspeakers. The response to it contains easily interpreted frequency domain and time domain information. The test is frequency selective, which helps to localize problems. Anechoic measurements can be made in normal rooms. The test results correlate with subjective sound perception.

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Figure 1.

Spectrum envelope of single sinusoid normalized to the sine wave frequency f_0 .

Figure 2.

Spectrum envelope of a 5-cycle tone burst normalized to the sine wave frequency f_0 . Note the slow low frequency roll-off.

Figure 3.

Spectrum envelope of a 5-cycle tone burst with approximately $\cos^2$ -shape.

Figure 4.

Time response of a second order high pass filter. (Frequency of poles: 1 KHz, $Q=2$).

Left: Response to rectangular window burst.

Right: Response to shaped burst.

Top traces are the rectified burst responses on a 10 dB/div. logarithmic amplitude scale. Bottom traces are the burst responses on a linear scale. Note the enhanced visibility of the burst envelope on a log scale.

A. 1 KHz input signal to the high pass filter (2 ms/div. horizontal).

B. Filter response at 1 KHz (2 ms/div. horizontal).

C. Filter response at 3 KHz (.5 ms/div. horizontal). Note the 1 KHz ringing of the rectangular burst response and its envelope variation.

Figure 5.

Frequency response of a second order high pass filter (poles at 1 KHz, $Q=2$).

A. Measured with a 5-cycle $\cos^2$ -shaped burst.

B. Steady-state frequency response. (Curves are offset 10dB).

Figure 6.

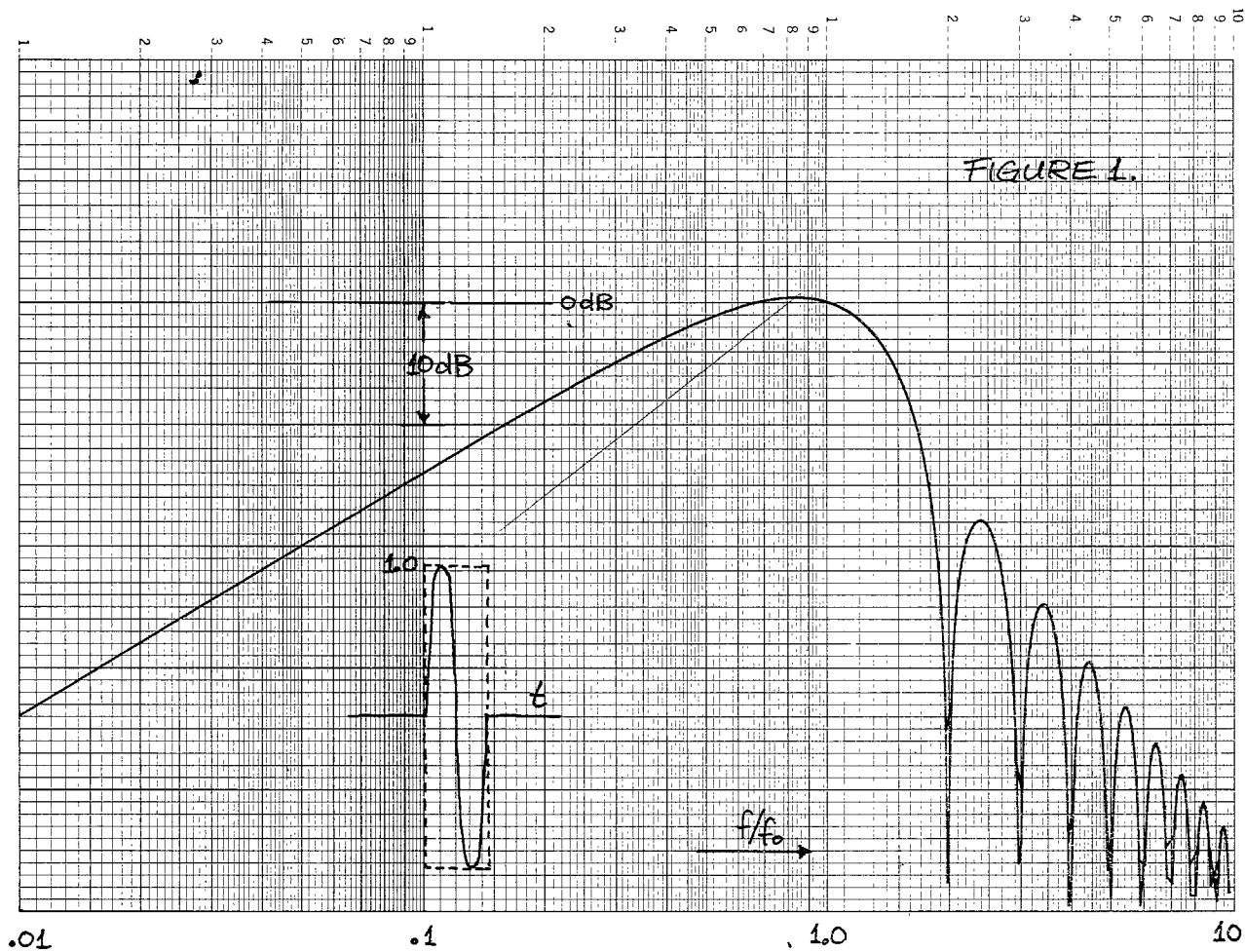
Time between the 5th zero-crossing of the drive signal and the high pass filter time response for a 5-cycle $\cos^2$ -shaped burst.

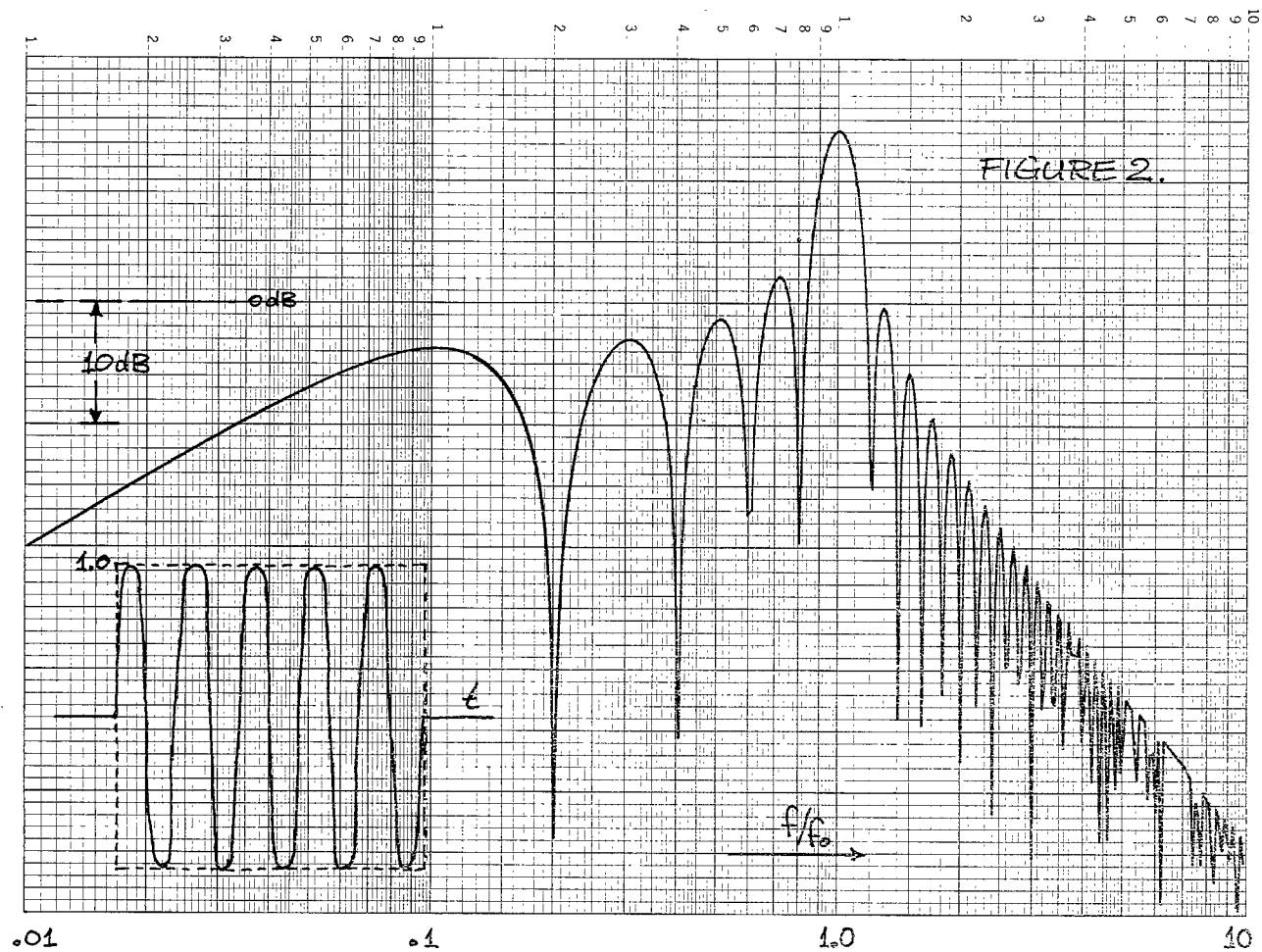
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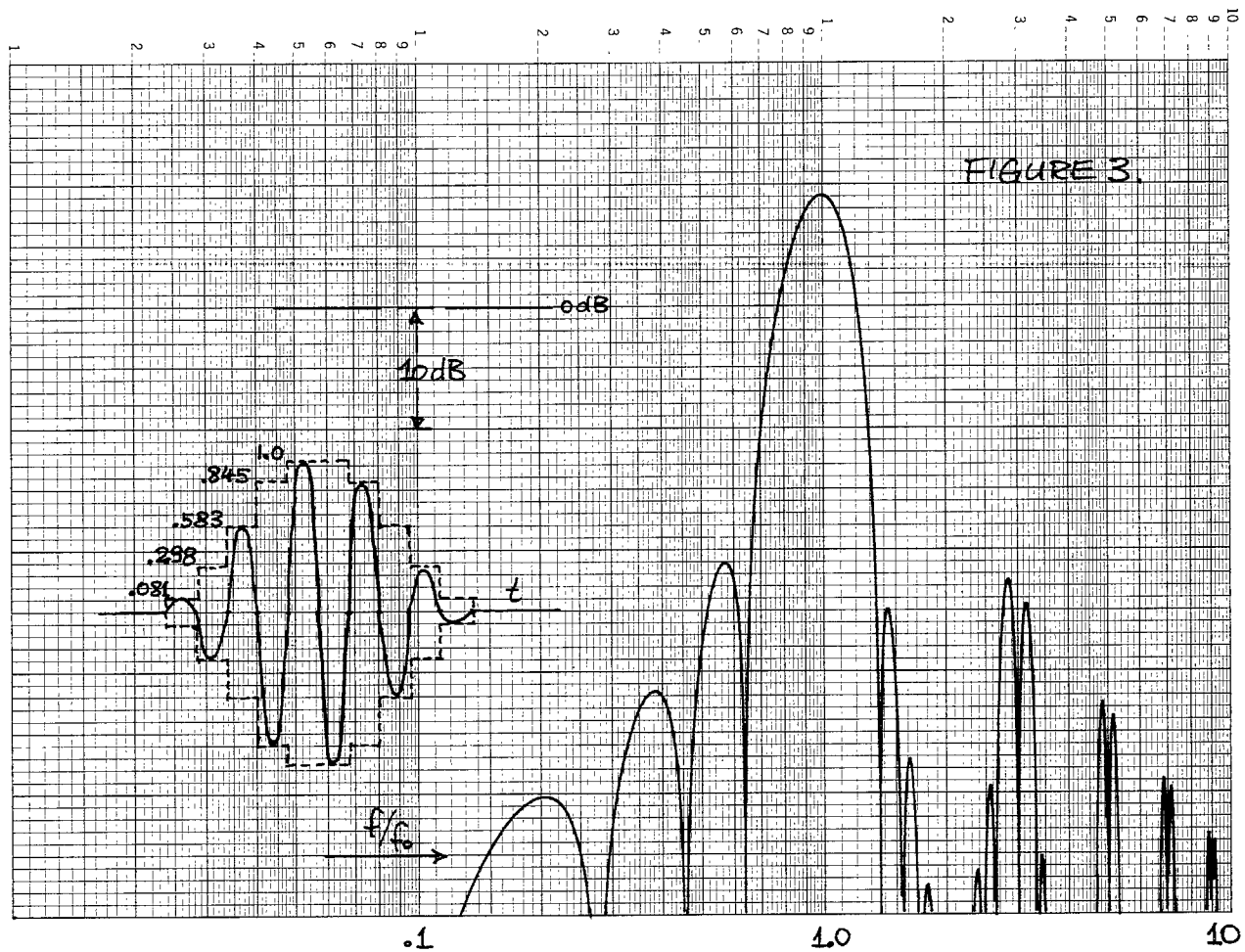
Block diagram of the test apparatus.

Figure 8.

Control signals for a programmable attenuator for generating a 5-cycle tone burst with approximately $\cos^2$ -shape.







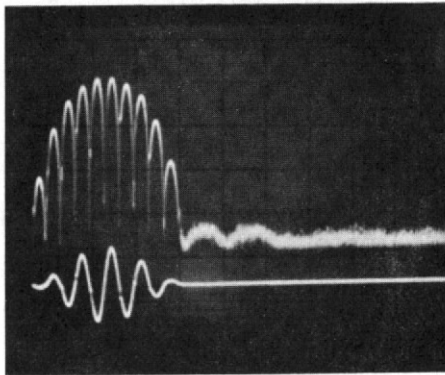
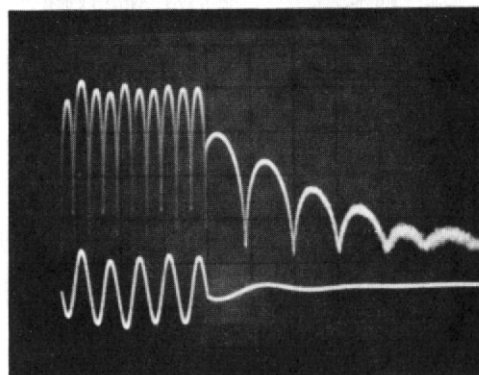
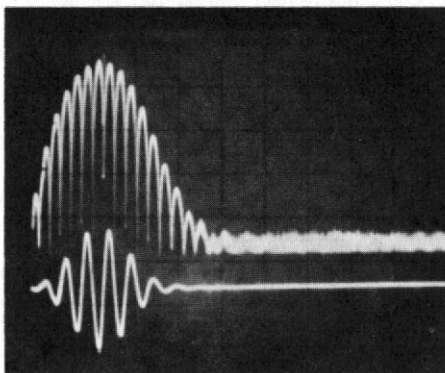
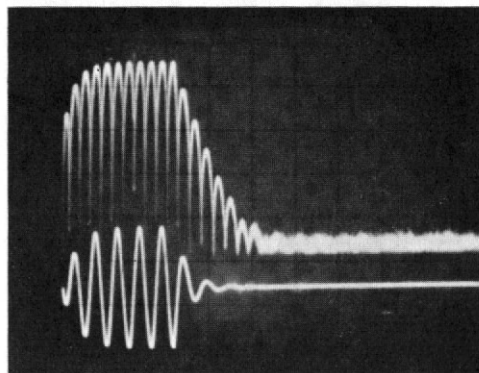
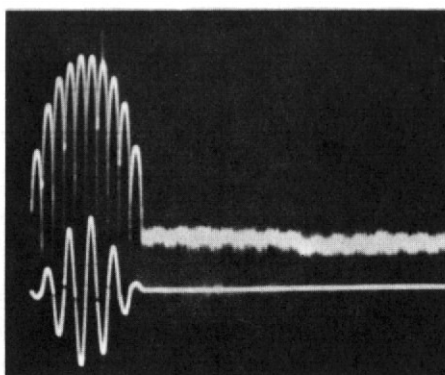
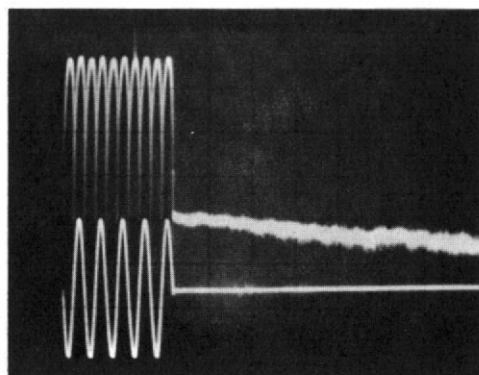
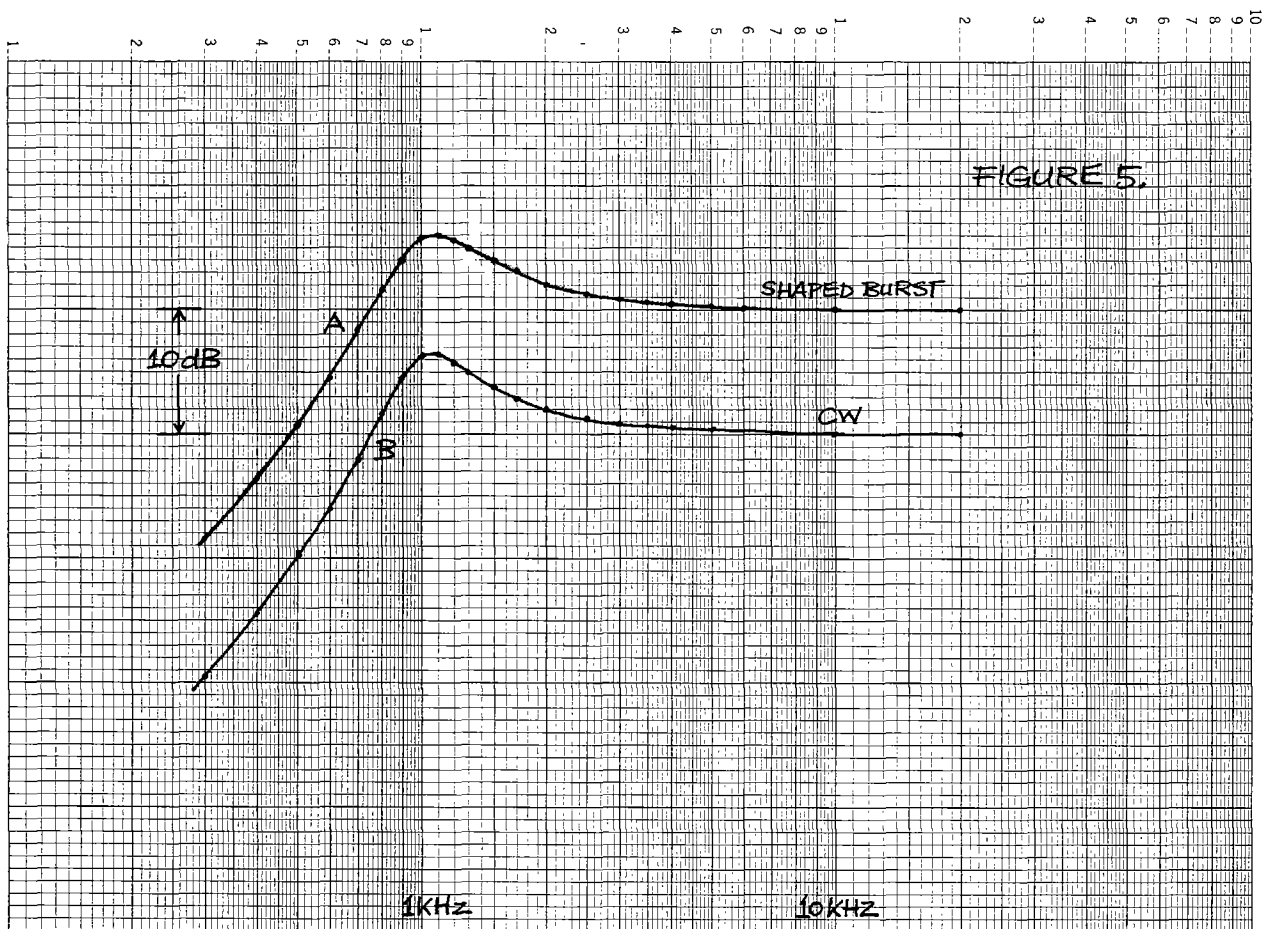
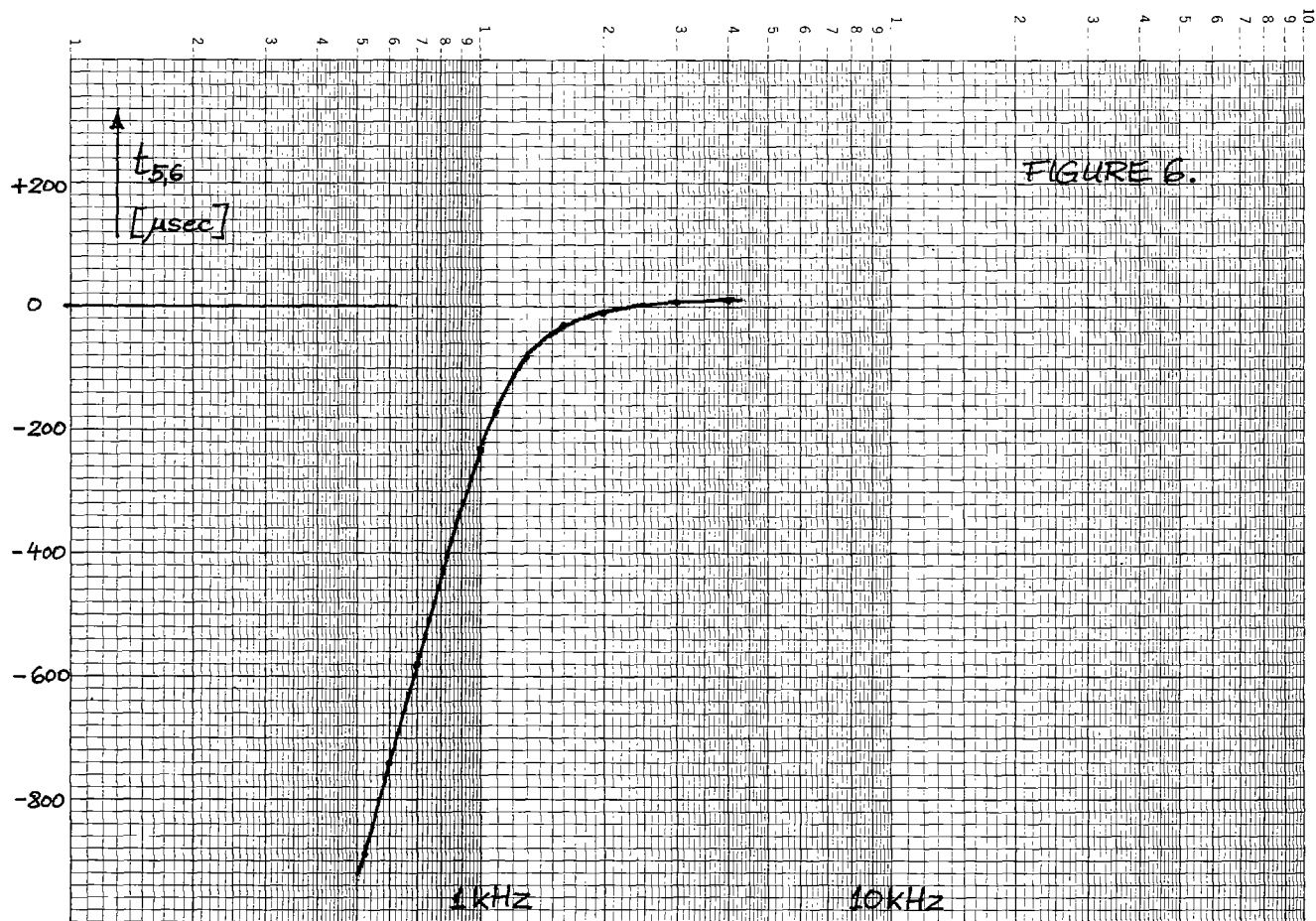


FIGURE 4.





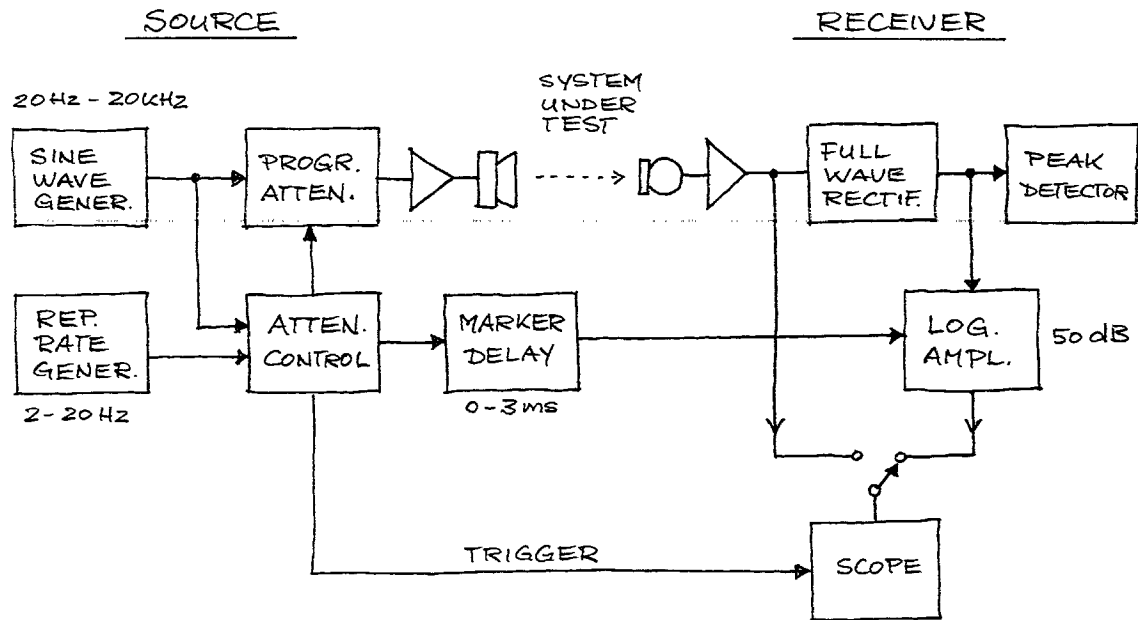


FIGURE 7.

FIGURE 8.

